

Self-applied ear-EEG for sleep monitoring at home

Kaare B. Mikkelsen^{1,*}, Yousef R. Tabar¹, Hans O. Toft², Martin C. Hemmsen²,
Mike L. Rank² and Preben Kidmose^{1,*}

Abstract—High quality sleep monitoring is done using EEG electrodes placed on the skin. This has traditionally required assistance by an expert when the equipment needed to be mounted. However, this creates a limitation in how cheap and easy it can be to record sleep in the subject's own home.

Here we present a data set of 120 home recordings of sleep, in which subjects use self-applied ear-EEG monitoring equipment.

We compare this data set to a previously recorded data set with both ear-EEG and polysomnography, which was applied by an expert.

Clinical relevance — On all tested metrics, self applied sleep recordings behaved the same as expert applied. This indicates that ear-EEG can reliably be used as a home sleep monitor, even when subjects apply the equipment themselves.

I. INTRODUCTION

Sleep is important to our health. We spend on average one third of our lives asleep, and the circumstances and quality of this sleep can be both affected by and have an effect on many different parts of our well being [1].

However, sleep is a delicate phenomenon to monitor: it changes from night to night, it is impacted by our surroundings, and it requires advanced instruments to monitor well. This has led to ongoing developments of new minimally invasive technologies for accurate sleep monitoring at home ([2]–[7]). As part of this, in Mikkelsen et al 2019 and 2021 ([3], [8]), we showed that dual-sided ear-EEG can be used as an alternative to full PSG, for healthy, young subjects. This is very valuable, in that it paves the way for ear-EEG-based sleep monitoring outside the clinic.

Here we explore the possibilities for home-applied ear-EEG for sleep monitoring. We supplied 10 subjects equipment for home monitoring, and had them apply the equipment at home. Each subject performed 12 successful home sleep recordings, yielding a total of 120 home recordings.

Throughout this report, we will compare this new home-applied data set (from 10 subjects) to the original expert applied data set (20 subjects). We will interchangeably refer to the old and new data sets as A and B or 'expert applied' and 'home applied'.

II. METHODS

A. Setup

The recording equipment consisted of 12 ear-EEG electrodes (Left ear: ELA, ELB, ELC, ELT, ELE, ELI, right ear: ERA, ERB, ERC, ERT, ERE, ERI), a single EOG

electrode (placed next to the right eye), a single-use sticker electrode (ground), a TMSi Mobita amplifier and a set of custom-molded silicone earpieces. See Figure 1 for an example of a person wearing the equipment, and previous ear-EEG literature [3] for an overview of the positions of the electrodes within the ears. The electrodes were dry-contact IrO electrodes [9]. Additionally, the subjects wore an 'Activeinsights' wristworn actigraph, however we will not discuss those data in this study.

We recruited 10 subjects among the original 20 subjects from the first study (mean age: 26 years, 7 women). The subjects were compensated monetarily for participation.

When comparing the two data sets, we have chosen to use all 20 original subjects for representing the 'expert applied' dataset, rather than only the 10 recurring subjects.

For all subjects, a significant time passed between the end of part A and the beginning of part B - at least 8 months, usually more than a year.



Fig. 1. Person sleeping while wearing the recording setup. Note the EOG electrode attached next to the eyebrow, as well as the adhesive electrode sticker placed on the neck. The bundle of wires leading into the ear are attached to the ear-EEG electrodes. Written permission was obtained from the person in the photograph.

Prior to their first home applied recording, each subject received brief instructions (about 10 minutes) on how to apply the electrodes, and how to turn on the amplifier. The subjects were allowed to receive help from any spouses or similar who might be available during application. We did not keep track of whether they took advantage of this.

Prior to each recording, the subjects were given a box containing the equipment, adhesive tape for any additional securing of wires or electrodes, a written instruction in how

¹ Department of Electrical and Computer Engineering, Aarhus University, Denmark

² T&W Engineering A/S, Denmark

*Corresponding authors: mikkelsen.kaare@ece.au.dk, pki@ece.au.dk

to perform recordings, and a sleep diary. In the diary they were asked to put down Recording Start, Lights Out, Wake Up and any events which they felt might have impacted the recording. Finally, in the morning, they had to answer some questions regarding the comfort and usability of the equipment.

Between recordings, the actigraph, sleep diary and amplifier were exchanged for 'fresh' copies.

The study was approved by the Central Denmark Region Committees on Biomedical Research Ethics (Ref. nr. 1-10-72-413-17) and the Danish Medicines Agency (ref. nr. 2017111085). Additionally, the study was registered with clinicaltrials.gov, with the protocol ID 'EarEEGSleep2018'. The study was monitored by the 'Good Clinical Practice' unit at Aarhus University. All participants provided written, informed consent and exclusion criteria were re-checked at every visit.

B. Signal quality

As part of the original study, we developed a pipeline for automatic artifact rejection, described in Mikkelsen et al 2019 [3]. We reused the pipeline in this study.

The pipeline also allowed us to keep an eye on the health of the hardware as the study progressed.

In our sleep scoring algorithm, the primary derivation used is the LR derivation: $LR = \text{avr}(\text{right ear electrodes}) - \text{avr}(\text{left ear electrodes})$. Given that this is well defined as long as each ear has at least one functional electrode, this is a very robust construction.

C. Sleep scoring based metrics

We also ran the recordings through a random forest based sleep scoring algorithm [3], using individualized models. This allows us a further two ways to compare the two datasets:

1) *Algorithm confidence*: We have previously shown a strong relationship between scoring quality and how 'confident' the algorithm is ([3], [10]). In this context, confidence for the i 'th night is measured as

$$c_i = \text{median}_j(\max_l(p_{lj})) \quad (1)$$

where p_{lj} is the conditional probability of the l 'th class for the j 'th epoch.

2) *Sleep metrics*: Given an automatic sleep scoring, we can calculate various standard metrics to describe how the subject has slept. In this study we focus on:

Total sleep: How many hours did the person sleep the entire recording.

Sleep latency: Time from 'Lights Out' to the first consecutive 5 minutes of sleep ('sleep onset').

Sleep efficiency: Percentage of time slept between 'sleep onset' and final awakening.

III. RESULTS

We successfully performed 120 self-applied home sleep recordings. On 9 occasions, a serious issue with the recording caused us to redo it. These issues were: earpiece fell out

during the night (2 times), subject forgot to turn on recorder (1 time), recording ended prematurely due to amplifier malfunction (6 times). Thus, in total, 129 recordings were attempted. In the analysis below, signal quality is calculated based on the 120 accepted recordings, while questions of subject comfort are based on the first 12 recordings from each subject (since the comfort is largely unaffected by the issues which lead to recording failure).

After each subject had finished his or her 12 recordings, we tested the health of all 13 electrodes. In total, we replaced 6 electrodes during the recordings. This is a very significant decrease relative to part A, where we replaced more than 100 electrodes over the course of 80 recordings (though, those recordings used a setup with 25 electrodes). We believe the cause of this significant drop is better control during electrode production (this change was implemented during part A) and the fact that the earpieces were not dismantled between recordings.

A. Data quality

When we look at signal quality, we find a generally lower level of artifact rejections, but also that inter-electrode differences are quite stable between the two datasets, see Figure 2.

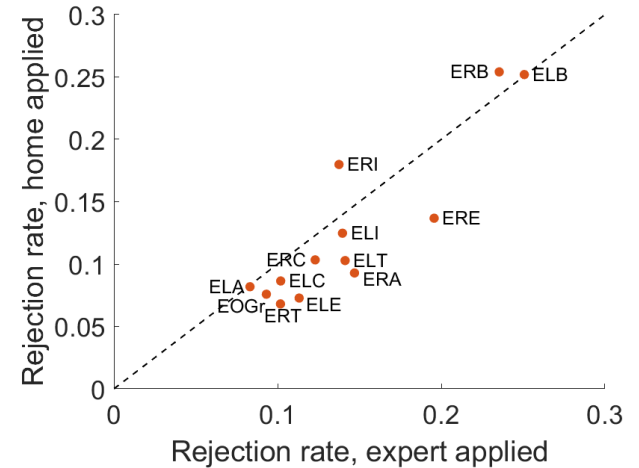


Fig. 2. Comparison between signal quality of individual electrodes in the two data sets. Dashed line indicates the identity. We see that most red dots are below the identity line, indicating a lower rate of data rejection in the home-applied data set.

When we consider the per-recording behavior of data rejection, again the two data sets are largely the same. Figure 3 shows the average missing samples in the LR composite derivation. The two distributions are very similar, meaning that the size of score-able portion of the recordings are the same.

B. Comfort

Regarding wearer comfort, we have two observables. The first one is the chosen delay between consecutive recordings - as the subject is in charge of scheduling each recording, we may interpret the time between recordings as an indicator

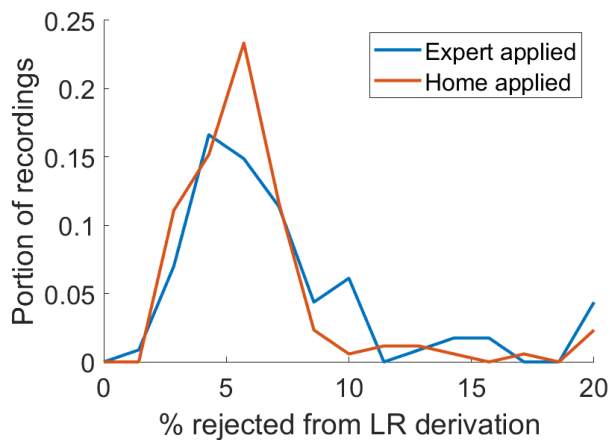


Fig. 3. Percentage of samples rejected from the 'LR' composite derivation, which is used for automatic sleep scoring. This is a direct measure of how large a portion of the recording can be used for automatic sleep scoring. We see that the distributions are very similar between the two data sets.

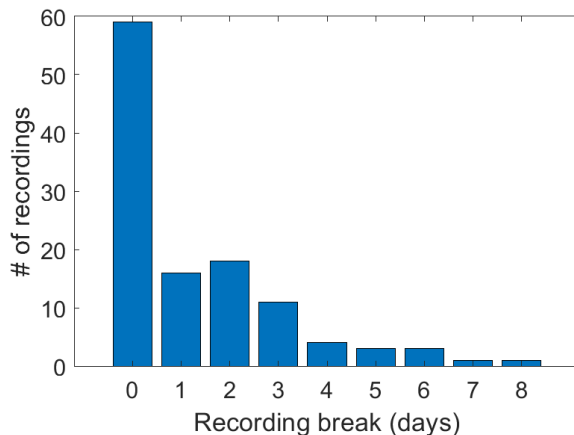


Fig. 4. Research subjects were free to schedule their recordings themselves, giving a picture of much rest they needed between recording nights. We see that very often, the subjects chose to do back-to-back recordings. In the preparation of this histogram, 3 intervals of more than 9 nights were removed. These were due to the covid-19 pandemic and vacations.

of whether the subject feels the need to 'rest their ears', or whether they are relatively unaffected. In Figure 4 is shown a histogram of the breaks between recordings. We see that in 59 out of 119 cases, the subject chose to do the next recording immediately after the previous recording.

The second observable is the answers given to the comfort questions in the sleep diary. In Figure 5 is shown average answers, during both the expert-applied and home-applied recordings. For all four questions, we see a clear trend of improvement during the first 4 recordings, but only little (if any) further improvement in the home-applied part.

An intriguing interpretation is that it is the PSG-portion which requires habituation, while the ear-EEG does not.

On one occasion, a subject woke up in the middle of the night with sore ears, and decided to sleep the rest of the night without the earpieces. This was night 11 out of 12, and the subject hypothesized that he had taped the wires too tightly.

This recording is included here.

C. Sleep scoring

In Figure 6 is shown the distribution of classifier confidence in the two datasets. We see that they are very similar, indicating that the automatic sleep scoring is likely also of the same quality.

Separately, Figure 7 shows distributions of the three sleep metrics. These are also very similar, indicating the change in circumstances did not change their sleeping patterns, and also supporting the hypothesis that sleep scoring in part B is of a similar high quality as in part A.

IV. DISCUSSION AND CONCLUSION

Compared to the expert-applied recordings, we obtained very high quality data. The ear-EEG equipment was at least as reliable in the home-applied setting (with the exception of the amplifiers, which seemed to be showing their age), and the research subjects were able to apply the earpieces just as well as the professionals. This matches observations that we made in the original study, where, on a few occasions, subjects had to re-apply the earpieces at home. However, the statistics in this study are much more solid.

We note a generally high level of comfort, and little sign of subject fatigue as the study wore on. In fact, there seems to be a slight increase in the perceived sleep quality at the end of the study.

It should be noted that our population comes with a slight bias: the subjects had to agree to participate in the home-applied study after the expert-applied study was finished. This means that the 10 subjects here do not represent the general population, but rather that part of the population which do not particularly mind sleeping with the ear-EEG. However, as we can see in Figure 5, there does not seem to be any appreciable increase in average comfort between the end of the expert-applied recordings and the beginning of the home-applied recordings. We interpret this to mean that the possible bias is insignificant.

In future work, we hope to develop the earpiece design more, moving towards a generic design, possibly without the external ground electrode. However, we believe that the present results are very promising regarding using ear-EEG as a tool for home-applied sleep monitoring at home.

ACKNOWLEDGMENT

This work was sponsored by the Innovation Fund Denmark, grant 7050-00007.

REFERENCES

- [1] R. B. Berry, R. Brooks, C. Gamaldo, S. M. Harding, R. M. Lloyd, S. F. Quan, M. T. Troester, and B. V. Vaughn, "AASM scoring manual updates for 2017 (version 2.4)," *Journal of Clinical Sleep Medicine*, vol. 13, no. 5, pp. 665–666, 2017.
- [2] K. B. Mikkelsen, D. B. Villadsen, M. Otto, and P. Kidmose, "Automatic sleep staging using ear-EEG," *Biomedical Engineering Online*, vol. 16, no. 111, Sep. 2017.

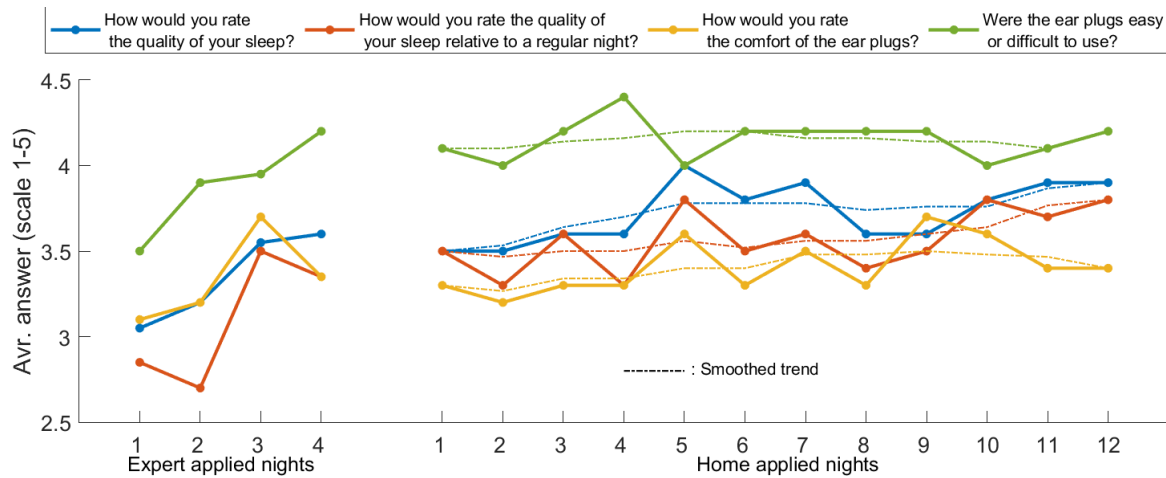


Fig. 5. Average answers to the four comfort questions. Dashed lines are moving averages, to make trends easier to see. We see a significant change in comfort during the four expert-applied nights, but only little change during the home-applied recordings.

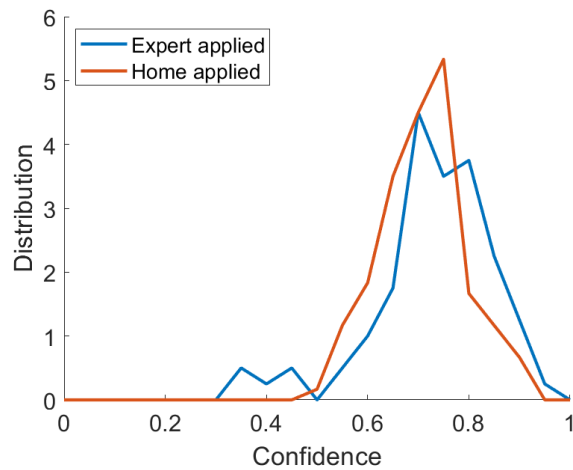


Fig. 6. Classifier confidence. This is correlated with sleep scoring accuracy. We see that the distributions are similar between data sets.

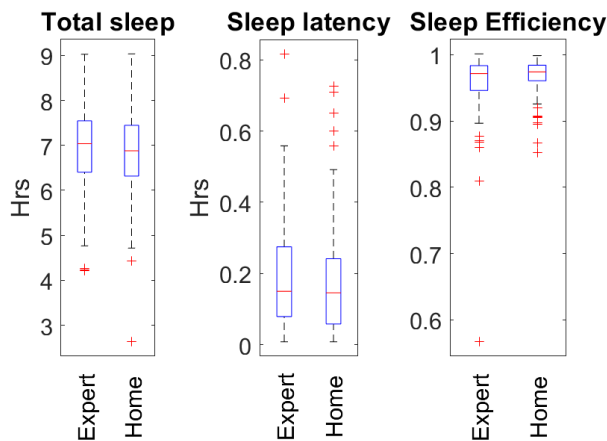


Fig. 7. Sleep metrics. Compares expert applied and home applied setups. We see that participants slept similarly with the two setups.

- [3] K. B. Mikkelsen, Y. R. Tabar, S. L. Kappel, C. B. Christensen, H. O. Toft, M. C. Hemmsen, M. L. Rank, M. Otto, and P. Kidmose, "Accurate whole-night sleep monitoring with dry-contact ear-EEG," *Scientific Reports*, vol. 9, no. 1, pp. 1–12, Nov. 14, 2019.
- [4] K. B. Mikkelsen, J. K. Ebajemito, M. A. Bonmati-Carrion, N. Santhi, V. L. Revell, G. Atzori, C. della Monica, S. Debener, D.-J. Dijk, A. Sterr, and M. de Vos, "Machine-learning-derived sleep–wake staging from around-the-ear electroencephalogram outperforms manual scoring and actigraphy," *Journal of Sleep Research*, vol. 0, no. 0, e12786, Nov. 13, 2018.
- [5] P. J. Arnal, V. Thorey, E. Debellemanniere, M. E. Ballard, A. Bou Hernandez, A. Guillot, H. Jourde, M. Harris, M. Guillard, P. Van Beers, M. Chennaoui, and F. Sauvet, "The Dream Headband compared to polysomnography for electroencephalographic signal acquisition and sleep staging," *Sleep*, vol. 43, zsa097 Nov. 12, 2020.
- [6] S. Kainulainen, H. Korkalainen, S. Siguroardóttir, S. Myllymaa, M. Serwatko, S. Siguroardóttir, M. Clausen, T. Leppänen, and E. S. Arnardóttir, "Comparison of EEG Signal Characteristics Between Polysomnography and Self Applied Somnography Setup in a Pediatric Cohort," *IEEE Access*, vol. 9, pp. 110916–110926, 2021.
- [7] S. Toften, S. Pallesen, M. Hrozanova, F. Moen, and J. Grønli, "Validation of sleep stage classification using non-contact radar technology and machine learning (Somnofy®)," *Sleep Medicine*, Mar. 6, 2020.
- [8] K. Mikkelsen, H. Phan, M. L. Rank, M. C. Hemmsen, M. De Vos, and P. Kidmose, "Sleep monitoring using ear-centered setups: Investigating the influence from electrode configurations," *IEEE transactions on bio-medical engineering*, vol. PP, Sep. 29, 2021. PMID: 34587000.
- [9] S. L. Kappel, M. L. Rank, H. O. Toft, M. Andersen, and P. Kidmose, "Dry-Contact Electrode Ear-EEG," *IEEE Transactions on Biomedical Engineering*, vol. 66, no. 1, pp. 150–158, Jan. 2019.
- [10] K. B. Mikkelsen, Y. R. Tabar, and P. Kidmose, "Predicting Sleep Classification Performance without Labels," in *2020 42nd Annual International Conference of the IEEE Engineering in Medicine Biology Society (EMBC)*, Jul. 2020, pp. 645–648.